

EGYPTIAN REAL ESTATE MARKET ANALYSIS

INTRODUCTION

The Egyptian real estate market has experienced rapid growth in recent years due to urbanization, population growth, and increasing investment demand. Understanding the factors that influence property prices is important for buyers, sellers, and investors. This study analyzes nearly 20,000 Egyptian property listings using statistical analysis and machine learning to identify the key drivers of residential property prices.

OBJECTIVE

This project aims to identify the most important factors affecting residential property prices in Egypt. The study examines the impact of property size, bedrooms, location, payment method, and property type using exploratory analysis, hypothesis testing, and linear regression.

METHODOLOGY

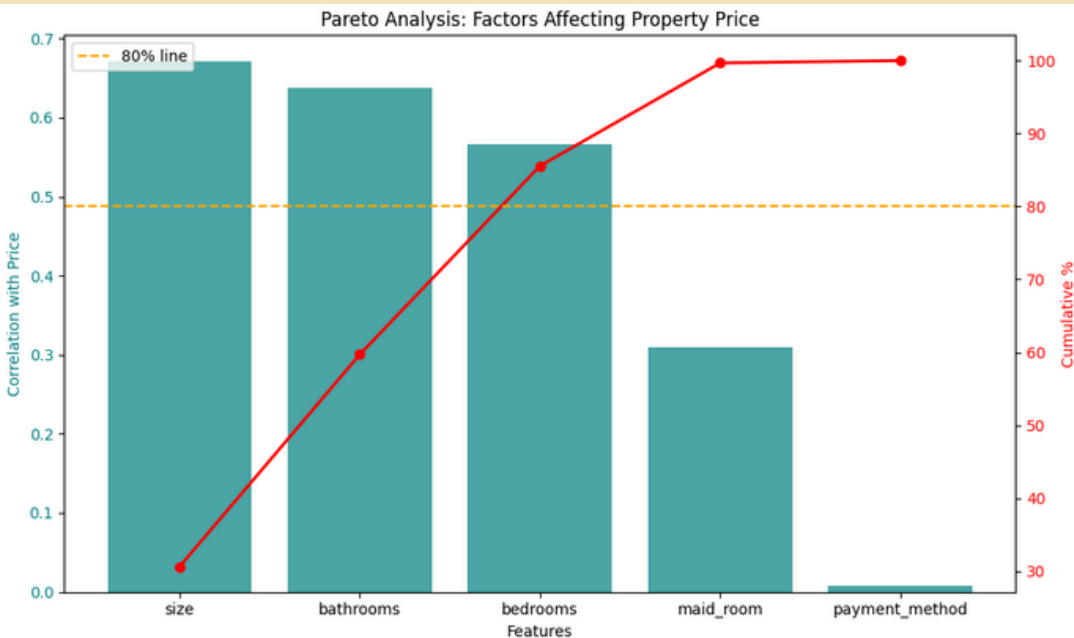
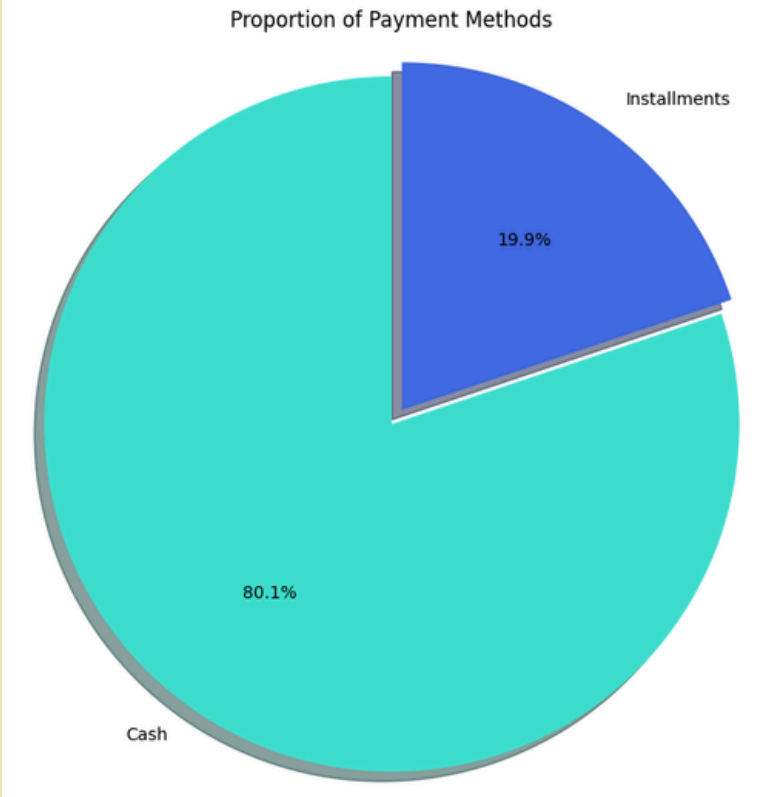
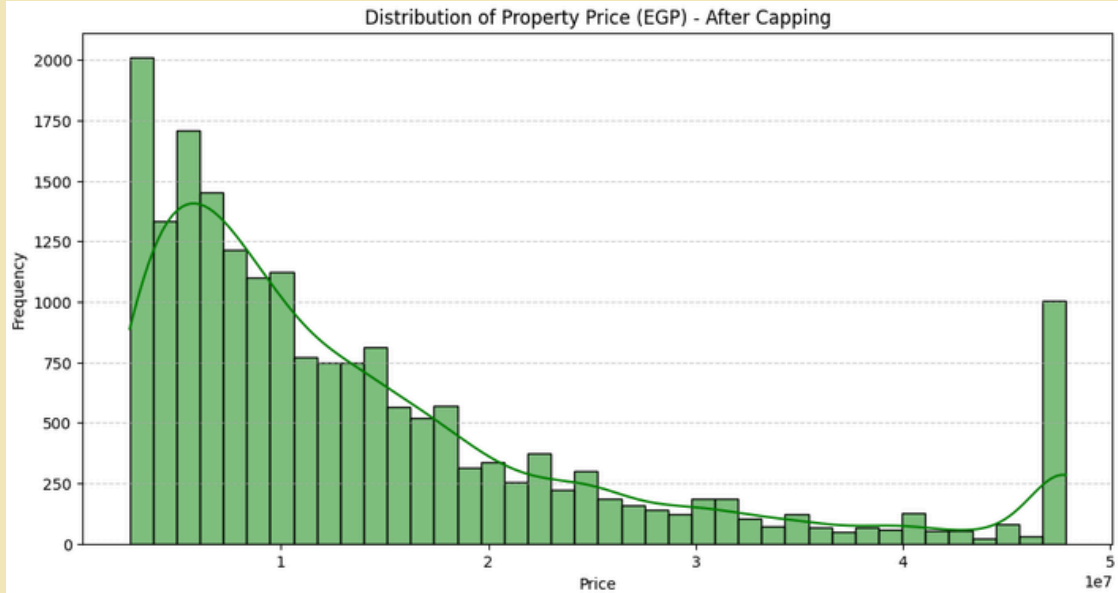
The dataset contains 19,925 Egyptian real estate listings collected from Kaggle. Data preprocessing included handling missing values, extracting numerical features, outlier capping, encoding categorical variables, and feature engineering. The analysis used:

- Exploratory Data Analysis (EDA)
- Correlation Analysis
- T-tests and ANOVA
- Linear Regression Modeling

DATA CLEANING AND PREPROCESSING

- Removed missing and duplicate values
- Handled outliers using capping techniques
- Converted categorical variables using encoding
- Extracted numerical features from textual columns
- Scaled and prepared data for regression analysis

EXPLORATORY DATA ANALYSIS



HYPOTHESIS TESTING

Hypothesis 1 — Property Size vs Price

Research Question

Does property size significantly affect property prices in Egypt?

Hypotheses

H0:Property size has no significant effect on property price

H1:Property size significantly affects property price

Hypothesis 2 — Bedrooms vs Price

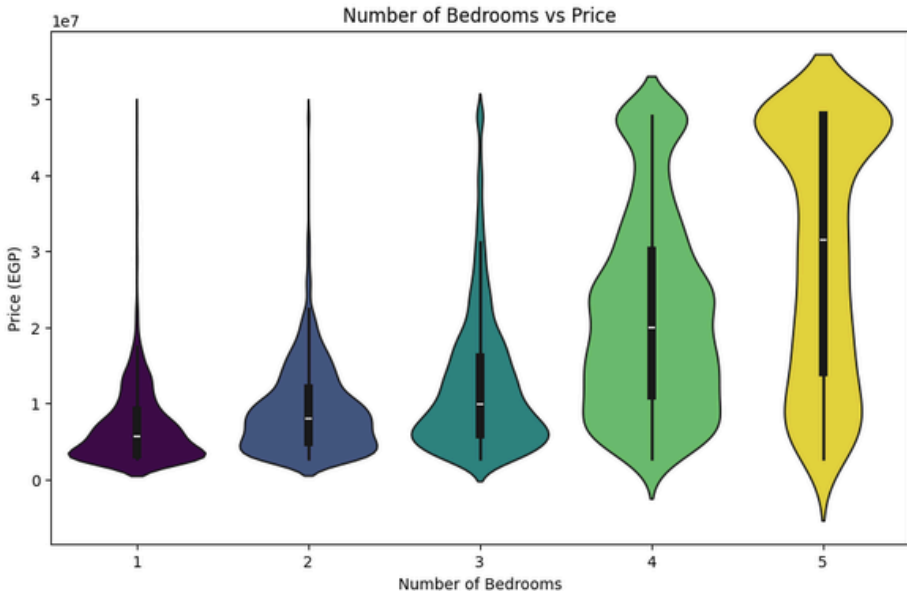
Research Question

Does the number of bedrooms significantly affect property prices?

Hypotheses

H0:The number of bedrooms has no significant effect on price

H1:The number of bedrooms significantly affects price



CONCLUSION

The hypothesis testing results showed that both property size and the number of bedrooms have a significant impact on residential property prices in Egypt. The analysis revealed a strong positive relationship between property size and price, while properties with more bedrooms consistently exhibited higher market values. Therefore, the null hypotheses for both tests were rejected in favor of the alternative hypotheses. Overall, this project successfully identified the main factors influencing the Egyptian real estate market using exploratory data analysis, statistical testing, and linear regression modeling. The findings demonstrated that physical property characteristics such as size, bedrooms, and property type are the strongest predictors of price, while some categorical factors showed weaker influence. The developed regression model achieved moderate predictive performance and provided valuable insights that can support buyers, sellers, investors, and real estate decision-making.

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